

The Physics of Music

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Music is essentially structured sound. The difference between music and noise is that music can be broken down into discrete frequencies which are related to each other by simple fractions, while noise is random and has a continuous frequency. In this essay I hope to explain how instruments produce notes and why every instrument has such a distinctive sound, as well as why some notes are harmonious when played together. It is impossible to understand the physics of music without first understanding the fundamental physics of sound waves.

1 Properties of sound waves

Sound waves transmit energy via vibrating particles in a medium such as air or water. Sound waves are longitudinal, meaning the propagation of the wave is parallel to the direction in which the particles in the medium vibrate. The vibrations of the particles in the medium leads to areas of high and low pressure, known as compressions and rarefactions respectively, as shown in figure 1. These changes in pressure can be detected by a sensitive membrane, for example the diaphragm of a microphone, and converted to a readable electric signal. A useful waveform to examine is one where pressure variation depends on a single frequency, resulting in a simple sine wave. This is known as a pure tone and has the equation, $y = A \sin 2\pi ft$, where y is the amplitude of a point on the wave, A is the maximum amplitude of the wave, f is the frequency of the wave and t is time. The shape of the waveform determines the properties of the sound produced. The amplitude of the wave determines how loud the sound will be, a higher amplitude leads to a louder sound. The frequency or pitch of the sound is determined by the equation, $v = f\lambda$, where v is wave speed, f is frequency and λ is the wavelength. Therefore, at constant wave speed, the pitch of the sound is inversely proportional to the wavelength. When two sound waves meet they interfere with each other. The interference adheres to the principal of linear superposition and depends on the phase difference between the waves. If the waves are out of phase a peak will align with a trough resulting in a waveform with a reduced amplitude. This is known as destructive interference. When two peaks or two troughs align, constructive interference occurs, leading to a louder sound.

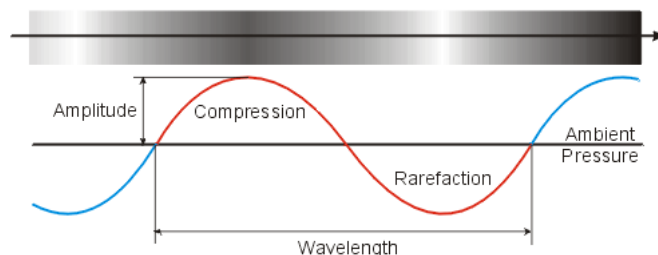


Figure 1: Pure tone waveform with one wavelength highlighted in red. The shaded bar represents varying pressure, darker meaning higher pressure.

2 How instruments produce notes

All acoustic instruments produce sound by oscillations, but they do this in a variety of different ways. Instruments can be grouped into loose categories including;

Idiophones - the entire instrument vibrates, for example a bell or cymbal

Membranophones - a stretched membrane vibrates, for example drums or singing

Chordophones - a stretched string vibrates, for example a guitar or piano

Aerophones - a column of air in the instrument vibrates, for example a trumpet or recorder

Acoustic instruments all function by creating standing waves when they are played. A standing wave is formed as the result of a wave either reaching the end of instrument and reflecting back the, interfering with itself. Let's take a guitar string as an example. Only waves with a whole number of wavelengths that fit on the string will constructively interfere, leading to large increases in amplitude known as resonance. Other frequency wavelengths will have their amplitudes reduced through destructive interference. The longest wavelength that fits the string is known as the fundamental wavelength. This is also normally the loudest and is perceived to be the pitch of the note. Other wavelengths that fit the string or column are known as overtones or harmonics. Overtone wavelengths are always a multiple of the fundamental. Points of minimum oscillation on the standing wave are known as nodes, while points of maximum oscillation are antinodes. A fixed string will have nodes at either end, as shown in figure 2.

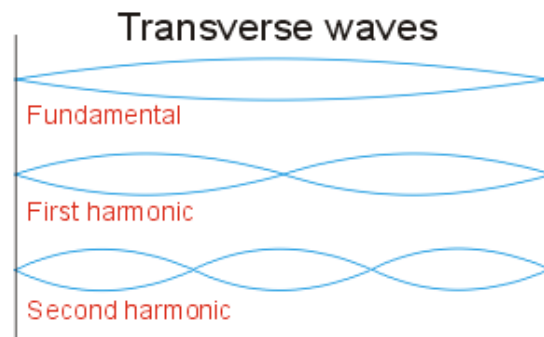


Figure 2: A diagram showing transverse waves on a string up to the second harmonic, the fundamental wave has an antinode in the middle of the wave.

An exception to producing sounds by oscillations are electrophones. These instruments use a capacitor and an inductor to form an oscillator circuit. This generates a sinusoidal electric signal which can be adjusted to any frequency. Several oscillators circuits can be combined at once and set to the frequencies of the overtones. The relative amplitude of these oscillator circuits can also be adjusted to produce a sound with the desired timbre. The circuit can be attached to a control source as well as an amplifier and speaker in order to create a fully programmable electronic instrument. All synthesizers function on this principle.

3 Why instruments sound different

Music is mainly polytonic; a mixture of pure tones played together to sound harmonious. Combining pure tones together to form a complex waveform is known as additive synthesis. A sound produced by multiple pure tones is still periodic but more complex than a simple sine wave. Any continuous complex function $f(t)$ can be broken down into an infinite series of sine and cosine functions in a process known as Fourier analysis, using the equation below,

$$f(t) = a_0 + \sum a_n \cos \left(\frac{2\pi n t}{T} \right) + \sum b_n \sin \left(\frac{2\pi n t}{T} \right)$$

Where t is time, T is the period $a_0 = \frac{1}{T} \int_0^T f(t) dt$, $a_n = \frac{2}{T} \int_0^T f(t) \cos \left(\frac{2\pi n t}{T} \right) dt$ and $b_n = \frac{2}{T} \int_0^T f(t) \sin \left(\frac{2\pi n t}{T} \right) dt$

Thus it can be seen that the waveform of any instrument is made up of multiple pure tones interacting with each other.

Some examples of complex waveform equations include a sawtooth wave, $y = \sum \frac{(-1)^{n+1}}{n} \sin nx$,

shown in figure 3, and a triangular wave, $y = \sum \frac{1}{(2n-1)^2} \cos(2n-1)x$, shown in figure 4.

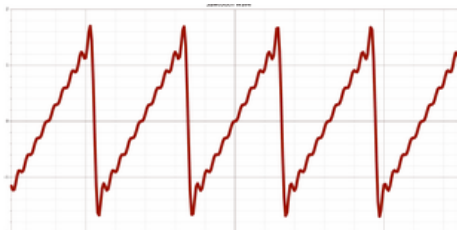


Figure 3: graph of a sawtooth wave

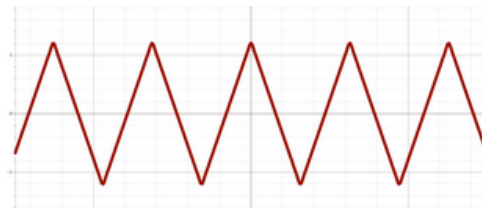


Figure 4: graph of a triangular wave

Different instruments produce a range of different waveforms. A pipe organ, for example, produces the waveform shown in figure 5, while a violin produces the waveform shown in figure 6. The extremely different sounds produced by both instruments are directly explained by the different shapes of their waveforms.

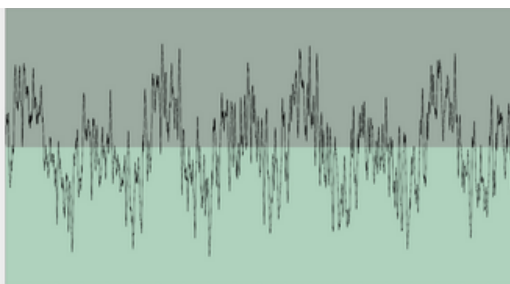


Figure 5: Pipe organ soundwave

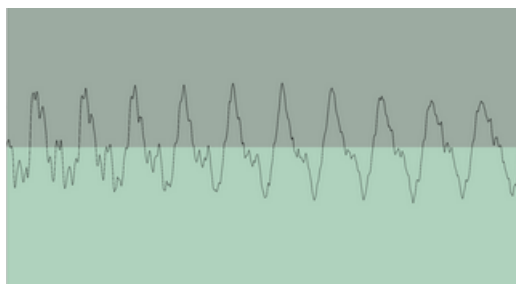


Figure 6: violin soundwave

4 Timbre

The relative amplitudes of the different overtones compared to the fundamental give a sound its timbre or quality, which can make it sound harsh or soft. Compare the overtones of a flute, shown in figure 7, and a recorder, shown in figure 8. These are two woodwind instruments that are essentially a tube with two open ends. This illustrates how the relative amplitude of the overtones differentiate each instruments sound.

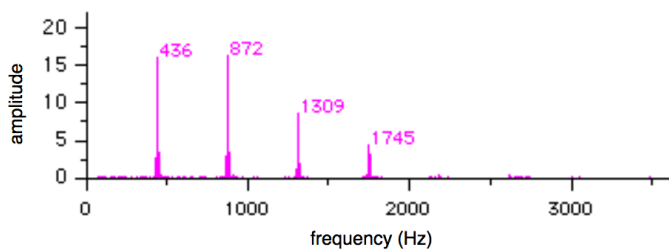


Figure 7: audio spectrum of a flute

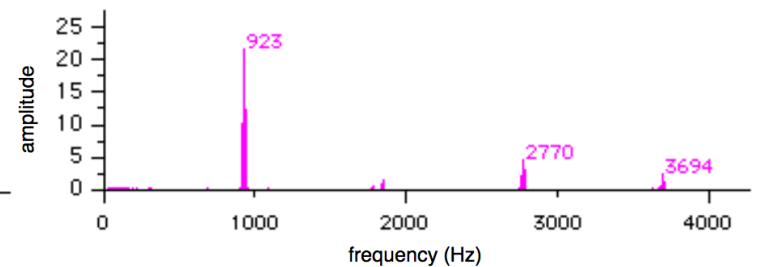


Figure 8: audio spectrum of a recorder

Notice the frequencies of the overtones of both a flute and recorder are multiples of the fundamental frequency. Comparing the two graphs we can see that the amplitude of the first overtone varies dramatically for each instrument, for a flute the amplitude is almost as large as the amplitude of the fundamental frequency. For a recorder, however, the amplitude of the first overtone is relatively small compared to the amplitude of the fundamental frequency.

5 Consonance and Dissonance

The ratio of any two pitches is known as an interval. Consonant intervals are judged to be pleasurable to listen to when played in unison as chords or in sequence as a melody. Sounds that are consonant usually have shared overtones. The more prime factors the numbers in the ratio of two pitches possess, the more dissonant that interval is. For example the interval of a perfect fifth, ratio $3/2$, is more consonant than the interval of a major seventh, ratio $15/8$ ($3 \times 5/2 \times 2 \times 2$). This is the basis of music theory and is used to construct chords and scales, the building blocks of music.

From this we can see that the physics of music is an important tool in understanding why some notes work together and others do not, essential for composing any song that would be enjoyable to listen to. The physics of music can also be used to construct electronic instruments that can be programmed to mimic any instrument or sound produced in nature, leading to a whole new world of possible musical adventures.

References

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Figure 1 from [4], *Method behind the music*

Figure 2-7 from [5], *Physics hypertextbook*

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